DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : PERIODIC ASSESSMENT - II

SET-A

SUBJECT : SCIENCE (086)

Time Allowed: 3 hours **M.M: 80**

Date : _____ **Name :** _____ **Class/Sec X**

General Instructions:

Read the following instructions carefully and follow them:
This question paper consists of 39 questions in 3 sections.

Section A is Biology, Section B is Chemistry and Section C is Physics.

All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

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Section – A

1. Dramatic changes in body features are associated with puberty that are mainly because of the secretion of..... (1)

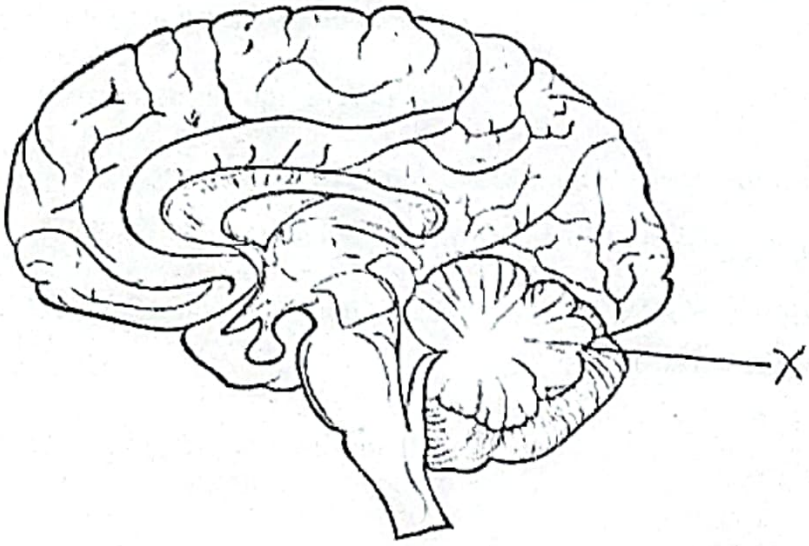
A. Oestrogen from testes and testosterone from ovary

B. Oestrogen from adrenal gland and testosterone from pituitary

C. Oestrogen from ovary and testosterone from testes

D. Oestrogen from pituitary and testosterone from thyroid

2. Observe the diagram below showing parts of the human brain. Identify the part labeled X and choose the correct function associated with it. (1)



A. X – Cerebellum; Maintains posture and balance

B. X – Medulla oblongata; Controls voluntary actions

C. X – Cerebrum; Regulates heartbeat and breathing

D. X – Hypothalamus; Coordinates reflex actions

3. Which of the following constitutes a food chain? **(1)**

A. Grass, goat, human

B. Goat, cow, human

C. Cow, grass, fish

D. Grass, fish, goat

4. The breakdown of pyruvate to give carbon dioxide, water and energy takes place in the: **(1)**

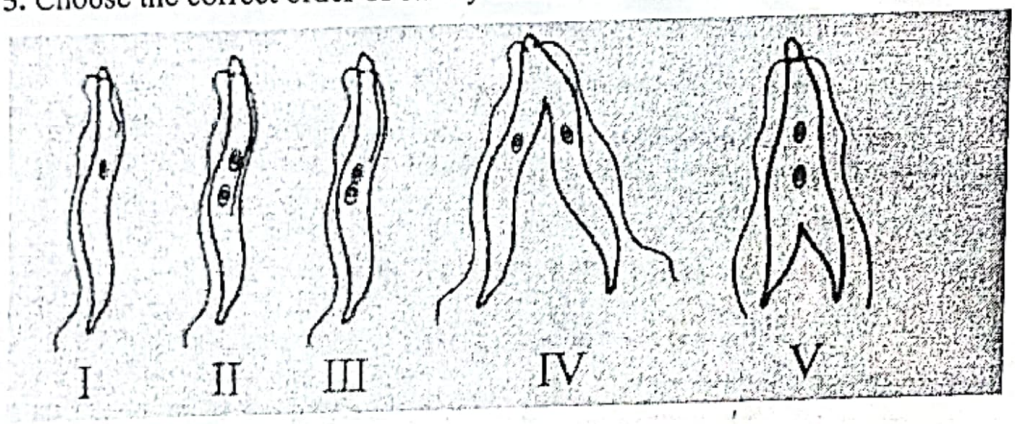
A. Cytoplasm

B. Mitochondria

C. Chloroplast

D. Nucleus

5. Choose the correct order of binary fission in *Leishmania*? **(1)**



A. I, II, III, IV, V

B. I, III, II, V, IV

C. I, III, V, II, IV

D. I, II, III, V, IV

6. In the given food chain, suppose the amount of energy at fourth trophic level is 5Kj, what will be the energy available at the producer level? **(1)**

Grass →Grasshopper→ Frog →Snake →Hawk

A. 5 kJ

B. 50 kJ

C. 500 kJ

D. 5000 kJ

7. Which of the following events in the mouth cavity will be affected if salivary amylase is lacking in the saliva? (1)

A. Starch breaking down into sugars.

B. Proteins breaking down into amino acids

C. Fats breaking down into fatty acids.

D. Absorption of vitamins.

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. **Assertion (A):** Garden is an artificial ecosystem.

Reason (R): Biotic and abiotic components are manipulated by humans in the garden. (1)

9. **Assertion (A):** *Amoeba* reproduces by binary fission.

Reason (R): All unicellular organisms reproduce only by asexual method. (1)

10. **State reason for the following:** (2)

A. Rings of cartilage are present in the trachea.

B. Plants look green in colour.

11. **Attempt either option A or B**

(A) Write any two ways by which bio-degradable wastes can affect the environment. (2)

OR

(B) What is biomagnification? Write one way by which the harmful chemicals enter into the food chain. (2)

12. **Attempt either option A or B**

A. Name the plant hormones responsible for the following functions: (2)

a. Apical dominance

b. promotes cell division

c. Germination of seeds

d. inhibits growth

OR

B. Write one example of each geotropism and chemotropism. Draw a neat labelled diagram of a potted plant showing positive and negative geotropism. (2)

13. A. Explain the role of UV radiation in producing ozone layer. Mention the reaction involved. (2)

B. Why is excessive use of CFCs a cause of concern? (1)

14. Rajesh observed a part of greenish black powdery mass on a stale piece of bread. Name the organism responsible for this and its specific mode of asexual reproduction. Draw a well labelled diagram of this organism and mention its non-reproductive and reproductive parts.(3)

15. Riya placed a potted sunflower plant near her window. Over the next few days, she noticed that the plant's stem slowly bent toward the sunlight. Curious, she rotated the pot, and again, the plant adjusted its direction toward the light. In her biology class, she learned that this behavior is a type of plant movement in response to external stimuli. Plants, although stationary, show movements in response to stimuli like light, gravity, touch, and chemicals.

Answer the following questions based on the case: (4)

Attempt either option A or B

(A) Why does the stem of a plant bend toward light during phototropism?

OR

(B) Differentiate between thigmotropism and chemotropism with examples.

(C) What type of plant movement is shown by the sunflower bending toward light?

(D) Name the plant hormone responsible for phototropism.

16. Attempt either option A or B

(A)

(a) Mention any two components of blood. (1)

(b) Write the movement of oxygenated blood in the body. (2)

(c) Write the function of valves present in between atria and ventricles. (1)

(d) Write one structural difference between the composition of arteries and veins. (1)

OR

(B)

(a) Name the organelle where photosynthesis takes place. (1)

(b) Write the overall balanced chemical equation for the process. (1)

(c) Name the three main events that occur during this process. Where do these events take place in the cell? (3)

$CaCO_3 + HCl \rightarrow CaCl_2 + H_2O$ CO_2

Section – B

17. A solution reacts with crushed egg-shells to give a gas that turns lime-water milky. The solution contains (1)

(a) NaCl (b) HCl (c) LiCl (d) KCl

18. Bleaching powder gives smell of chlorine because it (1)

(a) is unstable

(b) gives chlorine on exposure to atmosphere

(c) is a mixture of chlorine and slaked lime

(d) contains excess of chlorine

19. The colour of aqueous solution of zinc sulphate as observed in the laboratory is: (1)

a) Green b) Yellow c) Blue d) Colourless

20. What happens when dilute sulphuric acid is poured on silver plate:- (1)

(a) Silver sulphate is formed

(b) SO_2 gas is evolved

(c) No reaction takes place

(d) Hydrogen gas is evolved

$Ag + HCl \rightarrow$

21. Which of the following is the correct arrangement of the given metals in ascending order of their reactivity? Zinc, Iron, Magnesium, Sodium (1)

(a) Zinc > Iron > Magnesium > Sodium

(b) Sodium > Magnesium > Iron > Zinc

(c) Sodium > Zinc > Magnesium > Iron

(d) Sodium > Magnesium > Zinc > Iron

22. When an acid reacts with a metal carbonate or metal hydrogen carbonate, it gives the corresponding salt, water and releases which gas :- (1)

(a) O_2 (b) CO_2 (c) H_2 (d) N_2

23. Plaster of pairs is prepared from- (1)

(a) limestone (b) Slaked lime (c) quicklime (d) gypsum

Assertion and Reasoning Questions:

a) Both A and R are true, and R is the correct explanation of A

b) Both A and R are true, but R is not the correct explanation of A

c) A is true but R is false

d) A is false but R is true

24. Assertion (A): H_2CO_3 is a strong acid. (1)

Reason (R): Strong acid dissociates completely or almost completely in water.

25. In a beaker a solution of HCl is poured and an electric circuit containing bulb is placed systematically. What happens to the bulb and why? What will happen if HCl is replaced by NaOH? (2)

26. Attempt either option A or B (3)

(A) You are provided with three metals: Sodium, magnesium and copper. Using only water as the reactant, how will you identify them?

OR

(B) Give reason:

i. Electric wires are covered with rubber-like material.

ii. Iron grills are frequently painted.

iii. Gold ornaments retain their lustre even after several years of use.

27. For making cake, baking powder is taken. If at your home mother uses baking soda instead of baking powder in cake. (3)

(a) How will it affect the taste of the cake and why?

(b) How can baking soda be converted into baking powder?

(c) What is the role of tartaric acid added to baking soda?

28. Read the case study and answer the questions: **(4)**

The attractive force which holds various constituents (atoms, ions, etc.) together in different chemical species is called a chemical bond. Lewis postulated that atoms achieve the stable octet when they are linked by chemical bonds. In the formation of a molecule, only the outer shell electrons take part in chemical combination and they are known as valence electrons. The inner shell electrons are well protected and are generally not involved in the combination process. The bond formed, as a result of the electrostatic attraction between the positive and negative ions was termed as the electrovalent bond. The electrovalence is thus equal to the number of unit charge(s) on the ion.

Based on the above passage, answer the following questions:

i. Define Electrovalent bond.

ii. Show electrovalent bond formation in NaCl.

iii. Give any two important properties of Electrovalent compounds

OR

i. Show how $MgCl_2$ is formed.

ii. How is an ionic bond formed?

iii. Give an example of ionic compound commonly found in in your home.

iv. What types of atoms are most likely to form an ionic bond?

29. **Attempt either option A or B**

A. (i)Write balanced chemical equation for the following reactions. **(5)**

a. Ferrous sulphate crystals are heated in a dry boiling tube.

$FeSO_4 \xrightarrow{\Delta} Fe + SO_2 + SO_4$

b. A solution of potassium iodide is added to a solution of lead nitrate in a test tube.

$KI + PbNO_3 \rightarrow KNO_3 + PbI$

c. Barium chloride solution is mixed with copper sulphate solution.

$BaCl_2 + CuSO_4 \rightarrow BaSO_4 + CuCl_2$

d. Reaction of quick lime with water.

$CaO + H_2O \rightarrow Ca(OH)_2$

e. Reaction between Zinc and dil hydrochloric acid.

OR $Zn + HCl \rightarrow ZnCl_2 + H_2$

B. (i) A student prepared solutions of (i) an acid and (ii) a base in two separate beakers. She forgot to label the solutions, and litmus paper was not available in the laboratory. Since both the solutions are colourless, how will she distinguish between the two? (3)

(ii) What are strong and weak acids? (1)

(iii) In the following list of acids, separate strong acids from weak acids. Hydrochloric acid, citric acid, acetic acid, nitric acid, formic acid, sulphuric acid. (1)

SECTION - C

30. In which of the following, the image of an object placed at infinity will be highly diminished and point sized? (1)

a) Concave mirror only

b) Convex lens only

c) Concave mirror, convex mirror, concave lens and convex lens

d) Convex mirror only

31. How will the image formed by a convex lens be affected, if the upper half of the lens is wrapped with a black paper? (1)

a) The lower half of the inverted image will not be formed.

b) The image of upper half of the object will not be formed.

c) The brightness of the image will reduce.

d) The size of the image formed will be one - half of the size of the image due to complete lens.

32. **Assertion (A):** Red light signals are used to stop the vehicles on the road. (1)

Reason (R): Red coloured light is scattered the most so as to be visible from a large distance.

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

33. (A) The image of an object formed by a lens is of same size but inverted. If the object distance is 30 cm, calculate (1)

a. The distance between the object and its image.

b. Focal length of the lens.

(B) Draw a ray diagram to show the image formed in above case. (1)

OR

(A) A converging mirror forms a real image of height 4 cm, of an object of height 1 cm placed 20 cm away from the mirror. Calculate the image distance. 80 cm (1)

(B) What is the focal length of the mirror? ≈ 6.66 cm (1)

34. What is Tyndall effect? Give two examples where this phenomenon is observed. Give reason to explain this phenomenon. (2)

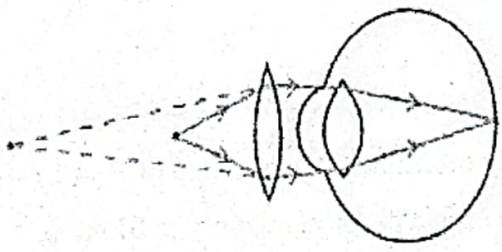
35. A concave mirror is used for image formation for different positions of an object. What inferences can be drawn about the following when an object is placed at a distance of 10 cm from the pole of a concave mirror of focal length 15 cm?

1. Position of the image
2. Size of the image
3. Nature of the image

Draw a labelled ray diagram to justify your inferences. (3)

36. Define power of a lens. The focal length of a lens is – 10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical center of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case? (3)

OR



The above image shows a corrective measure for a particular defect of vision. (3)

(i) Identify the defect of vision and state what kind of lens is used to correct this deficiency.

(ii) Draw and label a ray diagram that shows the defect of vision in the above case before correction.

37. (a) State two main causes of a person developing near - sightedness. With the help of a ray diagram, suggest how he can be helped to overcome his disability? (2)

(b) The far point of myopic person is 100 cm in front of the eye. Calculate the focal length and power of a lens required to enable him to see distant objects clearly. (1)

38. Read the following text carefully and answer the questions that follow:

Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors:

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

A. In which one of the above cases the mirror will form a diminished image of the object? Justify your answer. (1)

B. List two properties of the image formed in case 2. (1)

C. What is the nature and size of the image formed by mirror C? Draw ray diagram to justify your answer. (2)

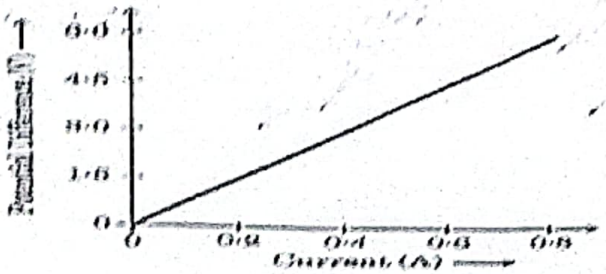
OR

An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case. (2)

39. Attempt either option A or B.

A. (a) State Ohm's Law. (1)

(b) Name and define the physical quantity determined by the slope of V - I curve given in the diagram. Use this graph to find the value of this physical quantity in SI units. (2)



(c) Define resistivity? What are the factors on which it depends? (2)

OR

B. a) Define resistance of a conductor. State the factors on which resistance of a conductor depends. (2)

b) Name the device which is often used to change the resistance without changing the voltage source in an electric circuit. (1)

c) Calculate the resistance of 50 cm length of wire of cross-sectional area 0.01 square mm and of resistivity $5 \times 10^{-8}\ \Omega\ m$ (2)